

Mathematics for Machine Learning: Homework 5

1. Using the definition of limit, prove that

$$\text{a) } \lim_{x \rightarrow 2} \frac{x^2 + 4x - 5}{x^2 - 1} = \frac{7}{3}, \quad \text{b) } \lim_{x \rightarrow +\infty} \frac{3x^2 + 5x - 7}{x^2 - 1} = 3.$$

2. Calculate the limit

$$\begin{aligned} \text{a) } \lim_{x \rightarrow 0} \frac{\operatorname{tg} x - \sin x}{\sin^3 x}, & \quad \text{c) } \lim_{x \rightarrow 0} \frac{\sqrt[5]{1 + \sin x} - 1}{\ln(1 + \operatorname{tg} x)}, \\ \text{b) } \lim_{x \rightarrow +\infty} \left(\frac{x^2 - 1}{x^2 + 1} \right)^{\frac{x-1}{x+1}}, & \quad \text{d) } \lim_{n \rightarrow +\infty} \left(\frac{\sqrt[n]{2} + \sqrt[n]{3}}{2} \right)^n. \end{aligned}$$

3. For which values of a the following function will be continuous on $\mathbb{R}$

$$f(x) = \begin{cases} \sin|x| - \ln|x|, & \text{if } |x| \geq 1 \\ ax^2 - 1, & \text{if } |x| < 1. \end{cases}$$

4. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function. Prove that there exists $c \in [0, 1]$ such that $f(c) = c$.
5. Let $f(x) = x(x-1)(x-2)(x-3)(x-4)$. Prove that the all roots of the equation $f'(x) = 0$ are lying inside the interval $(0, 4)$.
6. Find all extremum points of f

$$\begin{aligned} \text{a) } f(x) &= 2x^2 - x^4, & \text{c) } f(x) &= \cos x + \frac{1}{2} \cos 2x, \\ \text{b) } f(x) &= xe^{-x}, & \text{d) } f(x) &= e^x \sin x. \end{aligned}$$

7. Find maximum and minimum values of f in the given interval

$$\begin{aligned} \text{a) } f(x) &= 2^x, \quad x \in [-1, 5], & \text{c) } f(x) &= \sqrt{x} \ln x, \quad x \in (0, 1], \\ \text{b) } f(x) &= x^3 - 3x^2 + 6x - 2, & \text{d) } f(x) &= |x| + \frac{x^3}{3}, \quad x \in [-1, 1]. \\ & x \in [-4, 3], \end{aligned}$$

8. Calculate the integral

a) $\int \frac{dx}{x \ln x},$

c) $\int \frac{dx}{x^2 - x},$

b) $\int \frac{xdx}{\sqrt{x^2 + 1}},$

d) $\int \frac{dx}{\cos x}.$

9. Investigate convergence of the series

a) $\sum_{n=1}^{\infty} \frac{n^3 + 1}{n + 3} \sin \frac{1}{n^2 + 2},$

c) $\sum_{n=1}^{\infty} 2^n \left(\frac{n}{n+1} \right)^{n^2},$

b) $\sum_{n=1}^{\infty} \left(3^{\frac{1}{n}} - 1 \right) \sin \frac{\pi}{n},$

d) $\sum_{n=1}^{\infty} \frac{(3n)!}{(n!)^3 4^{3n}}.$